
FROM RATE SUPPRESSION TO OUTCOME STARVATION: EXACT FINITE-WIDTH CAUSAL THEORY FOR RECUR- RENT FEATURE COMPETITION

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ABSTRACT

Gradient starvation is usually discussed as weak or concentrated gradient flow, but a weak temporal feature can learn slowly for several distinct reasons: it may be intrinsically hard to remember, it may be helped by a stronger feature, or it may be causally suppressed by that feature. We separate these cases with a paired intervention in recurrent classifiers: a both-feature trajectory is compared with a same-initialization weak-only trajectory in which only the strong input channel is ablated. First, we derive exact finite-width signed-logit and response dynamics through the full sample empirical NTK and a response–logit cross-kernel, without a width limit or a frozen-kernel assumption. Second, we give a pathwise response-budget identity that applies to arbitrary numbers of rate sign changes and cleanly separates local rate suppression from outcome suppression; the familiar single-crossing “negative tail repays positive transfer” criterion appears only as a corollary. Causal starvation additionally requires that the weak-only counterfactual itself learns. Controlled dense-linear RNN experiments show positive weak-trajectory AUC gaps in every seed at learnable lags 0, 2, 4, while a reversed-cue control transfers and lag-8 cells are correctly left indeterminate. Nonlinear tanh and GRU audits show why rate crossings and causal labels should not be conflated. Finally, we derive Counterfactual Drift Correction (CDC), the unique uncapped minimum-norm local correction in the implemented parameter metric that reaches a weak-only drift target while exactly preserving instantaneous strong-response drift. Across eight dense-linear seeds, CDC reduces the weak AUC gap from 10.5155 to approximately zero while perturbing instantaneous strong drift by only 4.77×10^{-7} . The result is a finite-width causal account of recurrent feature starvation with explicitly scoped theory, evidence, and intervention guarantees.

1 INTRODUCTION

Cross-entropy training can make a model highly accurate while leaving useful predictive features underlearned. Gradient starvation formalizes one mechanism: once a dominant feature raises margins on many examples, those examples contribute little gradient, and weaker features can receive insufficient learning signal (Pezeshki et al., 2021). This mechanism sits near a broader literature on shortcut learning, simplicity bias, and spurious correlations, where models prefer easy predictive rules that can fail under shift (Geirhos et al., 2020; Shah et al., 2020; Sagawa et al., 2020; Liu et al., 2021; Kirichenko et al., 2023).

Recurrent networks make causal attribution harder. A weak temporal cue may be difficult simply because it is farther in the past. Shared recurrent parameters can also couple cues constructively: adding a strong cue can initially *increase* the weak-response learning rate before later suppressing it. Finally, a rate reduction is not yet an outcome: a feature can be temporarily suppressed and still finish ahead of its weak-only counterfactual. These distinctions are easy to lose if starvation is diagnosed from gradient magnitude or concentration alone.

We therefore ask a narrower causal question:

Does adding a strong predictive cue make an independently learnable weak cue fall behind the trajectory it would have followed without that strong cue?

We answer this with a paired intervention. The both-feature condition B and weak-only condition W share initialization, labels, samples, weak coordinates, noise, architecture, and optimizer; the only intervention is that the strong input channel is zeroed in W. We track the same weak response functional M_w in both models and distinguish:

1. **relative transfer:** $\dot{M}_w^B > \dot{M}_w^W$;
2. **relative rate suppression:** $\dot{M}_w^B < \dot{M}_w^W$;
3. **outcome suppression:** $M_w^B < M_w^W$;
4. **causal starvation:** outcome suppression plus independent weak-only learnability.

The last gate matters: if W itself cannot learn the weak cue, poor weak learning under B cannot be causally blamed on the strong cue.

Our theoretical starting point is exact and finite-width. For any differentiable response M_a and signed training logits r_i , full-batch logistic gradient flow obeys

$$\dot{M}_a = \frac{1}{n} \sum_i \langle \nabla M_a, \nabla r_i \rangle \sigma(-r_i). \quad (1)$$

The response–logit cross-kernel $\langle \nabla M_a, \nabla r_i \rangle$ measures how an update that improves example i changes the response of interest. This exact identity exposes why generic gradient concentration is insufficient for causal starvation: the weak-response rate depends on the *signed alignment* between the cross-kernel and the cross-entropy weights, and causal attribution additionally requires the matched counterfactual.

We then connect rates to outcomes without imposing a single rate crossing. Let

$$\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau), \quad d(\tau) = \Delta'(\tau). \quad (2)$$

The cumulative positive and negative variation of d exactly account for Δ at every time. This gives a general pathwise criterion for outcome suppression under arbitrary oscillations; the simple “negative tail exceeds earlier transfer area” rule is a one-crossing corollary rather than the main theorem. We deliberately present this as a causal accounting identity, not as a difficult analytic result. Its value is that it blocks a common logical error: a negative rate, or even a rate crossing, does not by itself establish that the strong feature caused a worse learned weak representation.

Finally, we use the response geometry to construct Counterfactual Drift Correction (CDC). A matched weak-only shadow supplies the weak-response drift that is missing under B. CDC adds the smallest Euclidean correction, in the fixed implemented parameter chart, that reaches this target while remaining orthogonal to the strong-response gradient. Its guarantee is local and exact: instantaneous strong-response drift is preserved, while finite-step deviation is second order in the step size. CDC is therefore best understood as a mechanism-level control experiment, not a drop-in practical optimizer.

Contributions.

- **Exact finite-width causal dynamics.** We derive universal response–logit cross-kernel dynamics and exact full sample signed-logit dynamics, with a closed $\dot{M} = Gg$ form only under explicitly stated response-coordinate realization assumptions.
- **A pathwise causal notion of starvation.** We separate transfer, rate suppression, outcome suppression, and weak-only learnability. A general positive/negative response-budget identity handles arbitrary rate sign changes; a single-crossing tail-area certificate follows as a corollary.
- **Controlled recurrent evidence.** Dense-linear RNNs show robust causal weak-trajectory deficits throughout the learnable region, while negative controls and nonlinear tanh/GRU audits expose cases where rate crossings, outcome crossings, and learnability disagree.
- **Theory-aligned counterfactual control.** CDC uniquely minimizes the local correction norm subject to weak-drift rescue and instantaneous strong-drift preservation in the implemented metric; experiments verify the protected drift to numerical precision.

2 RELATED WORK

Gradient starvation, shortcut learning, and simplicity bias. Pezeshki et al. (2021) derive gradient starvation under cross-entropy and propose Spectral Decoupling. Our question is narrower and causal:

whether adding a strong temporal feature makes a weak but independently learnable feature fall behind its matched ablation trajectory. Shortcut learning describes the broader tendency to rely on predictive but brittle cues (Geirhos et al., 2020), while simplicity bias studies optimization’s preference for simple predictive rules even when richer rules exist (Shah et al., 2020). These phenomena motivate our setting but do not supply the paired response-level causal criterion used here.

Spurious correlation robustness. Group DRO (Sagawa et al., 2020), Just Train Twice (Liu et al., 2021), and last-layer retraining methods such as DFR (Kirichenko et al., 2023) target worst-group performance when models exploit spurious features. Our work is not a competing robustness algorithm: the controlled synthetic intervention assumes explicit knowledge of the strong and weak channels, and CDC uses an even stronger matched-shadow information setting. We study the optimization mechanism and its causal attribution rather than claiming state-of-the-art robustness on natural benchmarks.

Learning dynamics in recurrent networks. Recent work derives low-dimensional learning dynamics for structured low-rank RNNs and shows that loss-invisible overlaps can retain training history (Ger and Barak, 2026). Statistical-physics analyses also characterize structure and disorder in task-trained recurrent circuits (Clark et al., 2026). Our main mathematical statements are different in scope: they are exact finite-width identities for a chosen response under full-batch gradient flow and do not require a mean-field closure. The empirical-NTK notation (Jacot et al., 2018; Lee et al., 2019) is useful for writing the exact signed-logit dynamics, but we do not treat a frozen kernel as exact for trained nonlinear networks.

Gradient surgery. PCGrad modifies conflicting task gradients (Yu et al., 2020); Bloop stabilizes an auxiliary projection with an exponential moving average (Hsieh et al., 2024). Projection itself is not novel here. CDC differs in the protected quantity—a theory-defined strong-feature response—and in its target, which comes from a matched weak-only causal shadow. Because this uses extra supervision and compute, we present CDC as a theory-aligned intervention rather than a universally deployable optimizer.

3 CAUSAL SETUP

3.1 PAIRED RECURRENT TRAINING

For condition $C \in \{B, W\}$, let $\theta^C(\tau) \in \mathbb{R}^P$ follow full-batch gradient flow on signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$:

$$L^C(\theta) = \frac{1}{n_C} \sum_{i=1}^{n_C} \log(1 + e^{-r_i^C(\theta)}), \quad \dot{\theta}^C = -\nabla L^C(\theta^C). \quad (3)$$

The two conditions share initialization and all data except that the strong channel is set to zero under W. Let M_s^C, M_w^C be common differentiable response functionals. Shared initialization gives $\Delta(0) = 0$ for

$$\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau), \quad d(\tau) = \Delta'(\tau). \quad (4)$$

Optimization time τ is distinct from sequence index.

The controlled dense-linear RNN is

$$h_{k+1} = Wh_k + Bx_k, \quad f(x) = c^\top h_T, \quad (5)$$

with $W_{ij}(0) \sim \mathcal{N}(0, g^2/N)$ and input/readout coordinates scaled as $\mathcal{N}(0, 1/N)$. We also audit tanh RNNs and GRUs. For nonlinear models, responses are common odd symmetric unit-probe contrasts; we do not assume that every nonlinear training logit lies in their two-dimensional span.

3.2 OUTCOME AND LEARNABILITY

Outcome suppression at time t means $\Delta(t) < 0$. Causal starvation additionally requires that the weak-only trajectory is demonstrably learnable within horizon H . Define the threshold-free weak-only reach

$$B_W(H) = \max_{0 \leq t \leq H} M_w^W(t), \quad (6)$$

and its complete threshold profile

$$\Lambda_W(\beta; H) = \mathbf{1}\{B_W(H) \geq \beta\}. \quad (7)$$

Thus a causal-starvation label at operating point β requires outcome suppression and $\Lambda_W(\beta; H) = 1$, while the entire β sensitivity of learnability is encoded by the single recorded quantity $B_W(H)$. The core response-gap and AUC results do not depend on β ; hitting-time and categorical causal labels do. We report the nonlinear counts at their preregistered $\beta = 0.5$ and do *not* interpret that single operating point as an architecture ranking.

Our threshold-free weak-trajectory summary is

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) d\tau, \quad (8)$$

computed by trapezoidal integration on the logged grid. Positive $\mathcal{A}_w$ means the weak-only trajectory is ahead on average over the horizon.

4 EXACT FINITE-WIDTH CAUSAL THEORY

4.1 UNIVERSAL RESPONSE-LOGIT DYNAMICS

Theorem 1 (Exact response-logit cross-kernel flow). *At every finite width and every optimization time at which M_a^C is differentiable,*

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i K_{ai}^C \sigma(-r_i^C), \quad K_{ai}^C = \langle \nabla M_a^C, \nabla r_i^C \rangle. \quad (9)$$

No width limit or two-mode realization is required.

Proof. Since $\ell'(r) = -\sigma(-r)$, Eq. (3) gives $\dot{\theta} = (1/n) \sum_i \sigma(-r_i) \nabla r_i$. Applying the chain rule to M_a yields Eq. (9). Full details and the corresponding signed-logit dynamics are in Appendix A.1.

Writing the signed-logit Jacobian $J \in \mathbb{R}^{n \times P}$, $\Theta = JJ^\top$, $c_a = J \nabla M_a$, and $q(r) = \sigma(-r)$ componentwise gives the exact finite-width system

$$\dot{r} = \frac{1}{n} \Theta(\theta) q(r), \quad \dot{M}_a = \frac{1}{n} c_a(\theta)^\top q(r). \quad (10)$$

The kernels may evolve arbitrarily. If the signed logits admit the exact parameter-independent realization $r_i = \sum_b z_{ib} M_b$, as in our zero-background-noise linear task, then

$$\dot{M}_a = \sum_b G_{ab} g_b, \quad G_{ab} = \langle \nabla M_a, \nabla M_b \rangle, \quad g_b = \frac{1}{n} \sum_i z_{ib} \sigma(-z_i \cdot M). \quad (11)$$

For nonlinear probes, Eq. (11) has an explicit residual; direct-autograd Eq. (9) is primary.

4.2 RECURRENT GEOMETRY AT FINITE WIDTH

For a channel-localized dense-linear mode $M_a = c^\top W^{\ell_a} b_a$, direct differentiation gives

$$G_{ab} = p_a^\top p_b + \mathbf{1}\{\text{ch}(a) = \text{ch}(b)\} q_a^\top q_b + \langle S_a, S_b \rangle_F, \quad (12)$$

where $p_a = W^{\ell_a} b_a$, $q_a = (W^{\ell_a})^\top c$, and

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (13)$$

Thus temporal separation enters the exact response geometry through matrix powers and recurrent-gradient paths rather than through a heuristic lag penalty.

4.3 PATHWISE RATE-TO-OUTCOME ACCOUNTING

We separate the mathematically exact accounting statement from the harder question of what dynamics generate a given sign pattern.

Proposition 2 (Pathwise response budget). *Let Δ be absolutely continuous on $[0, H]$, $\Delta(0) = 0$, and $d = \Delta'$. Define*

$$P(t) = \int_0^t [d(u)]_+ du, \quad N(t) = \int_0^t [-d(u)]_+ du. \quad (14)$$

Then, for every $t \in [0, H]$,

$$\Delta(t) = P(t) - N(t). \quad (15)$$

Hence strict outcome suppression occurs by H iff $N(t) > P(t)$ for some $t \leq H$, regardless of how many times d changes sign. A causal-starvation label at threshold β additionally requires $\Lambda_W(\beta; H) = 1$.

This is deliberately a bookkeeping proposition: its contribution is the causal separation it enforces, not proof difficulty. It immediately handles oscillatory or immediate-suppression trajectories that do not satisfy a transfer-first one-crossing pattern.

Corollary 3 (Single-crossing tail-area certificate). *Suppose there is $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ almost everywhere on (τ_d, H) . Let*

$$A_+ = \int_0^{\tau_d} d(u) du, \quad N_{\text{tail}}(t) = \int_{\tau_d}^t -d(u) du. \quad (16)$$

Then $\Delta(t) = A_+ - N_{\text{tail}}(t)$ for $t \geq \tau_d$, and strict outcome suppression occurs by H iff $N_{\text{tail}}(H) > A_+$. If weak-only learning also satisfies Eq. (7), this becomes a causal-starvation certificate.

The corollary blocks a specific false inference: a $+\rightarrow-$ rate crossing only marks a local maximum of Δ ; it does not force the weak response to fall behind. Appendix A.3 gives the proof and a quantitative response-crossing bound. Appendix A.4 additionally gives a rigorous initialization-local sufficient certificate for an early rate crossing/outcome deficit under an explicit curvature-smoothness bound; we do not use it to make empirical calibration claims in this draft.

5 CONTROLLED RECURRENT EXPERIMENTS

5.1 PROTOCOL AND IDENTIFICATION STRATEGY

Synthetic sequences contain a strong cue of scale ρ and a unit weak cue at controlled positions separated by lag $\Delta\ell$. Positive tasks align both cues with the label; a negative control reverses the strong cue. Both/weak-only data and initialization are paired exactly. Training is deterministic full-batch cross-entropy with zero weight decay.

The controlled design is intentional rather than a proxy for a benchmark leaderboard. It supplies two objects required by the causal theory that natural data rarely provides exactly: an identified weak response functional and a clean intervention that removes only the strong cue while leaving the matched weak signal unchanged.

The dense-linear grid uses width 96, $\rho \in \{1, 2, 4, 8\}$, $\Delta\ell \in \{0, 2, 4, 8\}$, 256 samples, 500 steps at learning rate 0.01, recurrent gain 0.5, $\beta = 0.25$, and eight seeds. The nonlinear audit uses width-32 tanh RNNs and GRUs at $\rho = 4$, lag 2, 256 samples, 1000 steps, learning rate 0.01, cue noise 0.1, and $\beta = 0.5$. For nonlinear models, direct-autograd cross-kernel drift is logged at every evaluation point.

5.2 DENSE-LINEAR RNNs: ROBUST CAUSAL DEFICIT IN THE LEARNABLE REGION

Figure 1 summarizes the dense-linear causal grid. At lags 0, 2, and 4, all 32/32 ρ -seed records per lag have positive weak AUC gaps. Averaged across ρ , the lag-wise gaps are 5.76, 8.93, and 6.11. At each fixed learnable lag, the gap increases monotonically over the measured ρ values. The reversed-cue control transfers in 8/8 seeds, with mean $\mathcal{A}_w = -4.038$ and 95% CI $[-4.498, -3.578]$.

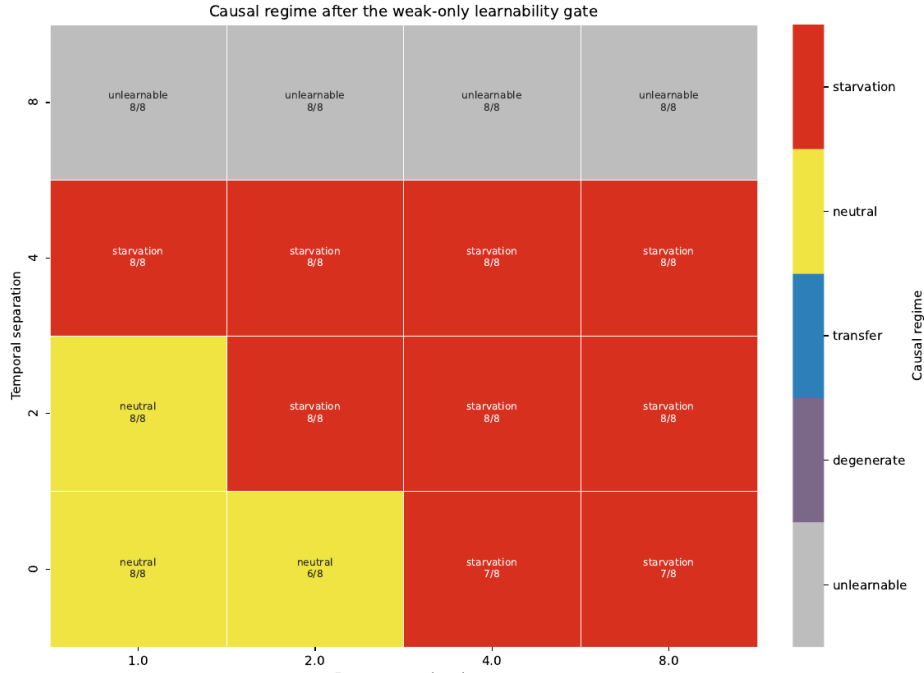


Figure 1: **Causal regimes in the controlled dense-linear RNN.** The weak-only learnability gate prevents long-lag memory failure from being mislabeled as gradient starvation. The AUC comparison itself is threshold-free; categorical labels use the preregistered operating point.

Table 1: **Nonlinear causal audit** at $\rho = 4$, lag 2, eight seeds. Counts are reported at the preregistered $\beta = 0.5$ learnability operating point; we do not infer an architecture ranking from this threshold.

Model	drift cross	response cross	causal	mean τ_d	mean τ_r
Tanh RNN	3/8	3/8	3/8	0.194	0.389
GRU	5/8	5/8	0/8	3.846	7.200

Every lag-8 cell fails weak-only learnability and is therefore left indeterminate rather than called starvation. Final accuracy is 1.0 throughout, emphasizing that average predictive performance does not reveal the weak-feature deficit.

The exact dense-linear geometry is independently checked against autograd with zero recorded relative error, and the maximum projected-identity error for Eq. (11) is 2.86×10^{-6} . These are implementation checks of analytic identities, not empirical proofs.

5.3 NONLINEAR AUDIT: RATE EVENTS ARE NOT CAUSAL LABELS

Table 1 reports the final direct-autograd nonlinear protocol. Tanh has a transfer-first drift crossing, later response crossing, and $\beta = 0.5$ learnability-gated causal certificate in 3/8 seeds. The remaining five seeds are rate-suppressed from initialization and therefore do not instantiate the preregistered transfer-first certificate. GRU has drift and response crossings in 5/8 seeds, but $\Lambda_W(0.5; H) = 0$ in all eight seeds, so the preregistered causal count is 0/8.

The point of this audit is logical, not comparative: response crossings can occur without passing the learnability gate, and immediate suppression need not follow a transfer-first pattern. The general pathwaywise Proposition 2 covers both cases; the stricter one-crossing certificate is retained only because it was preregistered for the reported nonlinear counts. Figure 2 shows the exact paired weak-drift difference; projected two-response decompositions are diagnostics only.

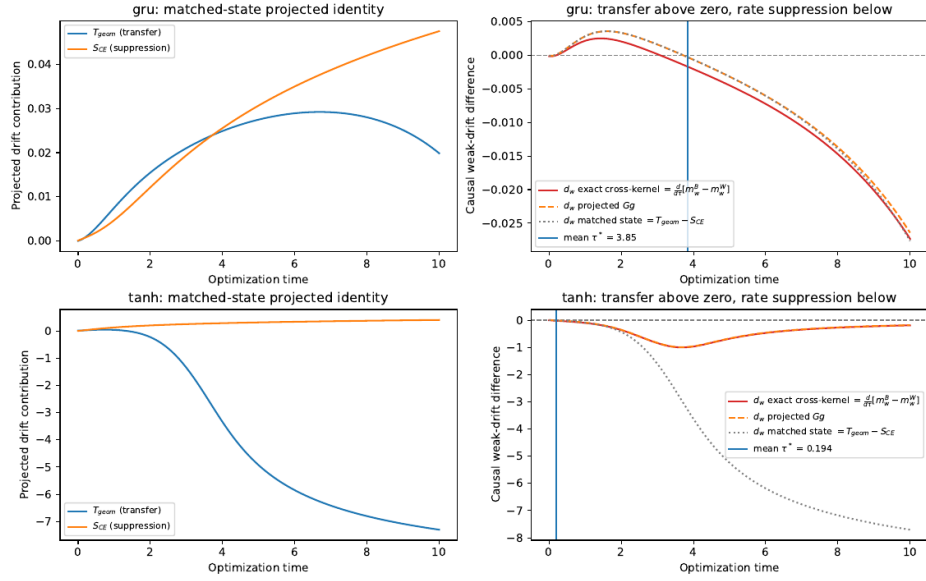


Figure 2: **Theorem-aligned nonlinear trajectories.** Direct-autograd paired weak drift is primary. Projected quantities are shown separately because $\dot{M} = Gg$ is not exact for the nonlinear probe responses.

6 COUNTERFACTUAL DRIFT CORRECTION

The theory identifies a response-level control objective: restore missing weak-response drift without changing the instantaneous strong-response drift. At a both-feature state θ , let

$$a = \nabla M_s(\theta), \quad p = \nabla M_w(\theta), \quad v = -\nabla L(\theta), \quad (17)$$

and let the matched weak-only shadow supply target F_w^W . Set $F_w = p^\top v$, $\delta = [F_w^W - F_w]_+$, and

$$q = \begin{cases} p - \frac{p^\top a}{\|a\|^2} a, & a \neq 0, \\ p, & a = 0, \end{cases} \quad \alpha = \begin{cases} \delta / \|q\|^2, & \delta > 0, q \neq 0, \\ 0, & \delta = 0. \end{cases} \quad (18)$$

The corrected velocity is $v' = v + \alpha q$. A positive deficit with $q = 0$ is infeasible.

Theorem 4 (CDC local guarantees). *In the fixed implemented Euclidean/Frobenius parameter chart, the feasible uncapped correction in Eq. (18) satisfies:*

1. $a^\top v' = a^\top v$ exactly (instantaneous strong-drift preservation);
2. $d^* = \alpha q$ is the unique minimum-Euclidean-norm correction satisfying $a^\top d = 0$ and $p^\top (v + d) \geq F_w^W$;
3. if ∇M_s is L -Lipschitz on a convex neighborhood containing both step segments, then

$$|M_s(\theta + \eta v') - M_s(\theta + \eta v)| \leq \frac{L\eta^2}{2} (\|v'\|^2 + \|v\|^2). \quad (19)$$

The theorem is deliberately local and coordinate-dependent. It does not imply final strong-response retention, reparameterization invariance, or practical deployability without the matched shadow.

We evaluate eight methods on the dense-linear starvation cell $(\rho, \Delta\ell) = (4, 2)$ with width 96 and eight paired seeds. All five shadow-based rescue methods reduce ERM's weak AUC gap of 10.5155 to approximately zero. CDC's distinctive empirical property is the theorem-matched constraint: its maximum instantaneous strong-drift perturbation is 4.77×10^{-7} , versus 1.48×10^{-1} – 3.75×10^{-1} for the alternative shadow constraints. CDC does *not* have the best final M_s ; this is consistent with a one-step local guarantee rather than a trajectory guarantee.

Table 2: **Dense-linear mitigation ablation**, eight paired seeds. Shadow methods use a matched weak-only model; CDC’s established distinction is protected instantaneous drift, not uniquely superior rescue.

Method	weak AUC gap	max strong-drift change	final M_s
ERM	10.5155	–	1.364
Spectral Decoupling	9.0229	–	0.829
CDC	−0.0005	4.77×10^{-7}	1.042
Unconstrained rescue	−0.0007	1.48×10^{-1}	1.295
PCGrad-style	−0.0007	1.48×10^{-1}	1.295

The full eight-method ablation is in Appendix D. PCGrad- and Bloop-style arms are repository implementations motivated by the cited algorithms, not independently audited replicas.

7 DISCUSSION AND SCOPE

What is novel. The elementary chain rule and the integral identity are not claimed as isolated mathematical novelties. The contribution is the *causal finite-width formulation*: a response-specific cross-kernel determines weak-feature drift; a matched ablation isolates the strong feature’s causal effect; pathwise accounting separates rate events from outcomes under arbitrary sign changes; and the same response coordinates define a constrained intervention. This differs from generic gradient-concentration diagnostics and from equilibrium or low-dimensional recurrent closures.

Why GSI is diagnostic rather than definitional. Legacy GSI statistics summarize gradient energy, margin dispersion, alignment, rank, or sigmoid-weight concentration. GSI-5 is directly related to the cross-entropy weights $q_i = \sigma(-r_i)$, but the exact weak-response drift is $c_w^\top q/n$. None of the five indices contains the signed response-specific alignment, the matched weak-only counterfactual, or the outcome accounting in Proposition 2. We therefore retain GSI-5 only as a supporting mechanism diagnostic.

Synthetic scope and external validity. All citable results are controlled synthetic experiments. This is a limitation for broad impact, but it is also what makes the causal intervention exact: strong-channel removal and weak-response identification are known by construction. A natural-data benchmark would require a defensible response coordinate and an ablation that does not change unrelated content. We make no real-data robustness claim in this draft.

Threshold and nonlinear scope. The central exact dynamics and the AUC gap do not depend on β . Only hitting-time and categorical causal labels do. The nonlinear table therefore reports a single preregistered operating point and is used to demonstrate logical distinctions, not to claim that tanh is more or less prone to starvation than GRU in general.

Selection and reproducibility. Superseded exploratory runs used mathematically different response/scaling conventions and are excluded wholesale from citable claims; we do not pool evidence across definitions. The corrected dense-linear and theorem-aligned nonlinear results use the final definitions. For double-blind review, all code, resolved configurations, tests, and citable artifacts should be packaged in an anonymous supplementary archive rather than linked to a public identity. Additional provenance details are in Appendix F.

8 CONCLUSION

A weak gradient is not enough to establish gradient starvation. In recurrent learning, a strong feature can transfer learning, suppress the weak-response rate, or coexist with a weak cue that is itself unlearnable. We provide a finite-width causal account that separates these cases: exact response–logit dynamics identify the response rate, matched ablation identifies the causal comparison, pathwise response budgets distinguish rate events from outcome suppression, and a learnability gate prevents ordinary memory difficulty from being mislabeled as starvation. CDC then controls the same

response-level deficit while exactly preserving instantaneous strong drift. The result is deliberately narrower than a universal theory of recurrent optimization, but every central claim is stated at the level the mathematics and experiments actually support.

REPRODUCIBILITY STATEMENT

Complete proofs, exact model definitions, causal-label rules, censoring conventions, hyperparameters, and the full CDC ablation are included in the appendix or anonymous supplementary package. The implementation contains executable checks for the exact dense-linear geometry, response-flow identities, pathwise certificate logic, and CDC algebra. The reported confirmatory experiments use fixed paired seeds and resolved configurations; superseded exploratory runs are not used for citable claims.

AI USE STATEMENT

Generative AI tools were used for literature discovery, hypothesis and methodology discussion, formulation and checking of mathematical claims and proofs, implementation/code-review assistance, interpretation of recorded results, and manuscript drafting/editing. AI-produced text or reasoning was not treated as empirical evidence: mathematical statements were checked against explicit derivations and executable theorem contracts, and empirical statements were checked against recorded experiment artifacts. The authors take responsibility for all final claims, proofs, citations, code, and experimental results.

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A PROOFS AND ADDITIONAL THEORY

A.1 PROOF OF THEOREM 1

For $\ell(r) = \log(1 + e^{-r})$, $\ell'(r) = -\sigma(-r)$. Therefore

$$\dot{\theta}^C = -\nabla L^C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \nabla r_i^C. \quad (20)$$

The chain rule gives

$$\dot{M}_a^C = \langle \nabla M_a^C, \dot{\theta}^C \rangle = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C), \quad (21)$$

which proves Eq. (9). Writing $\dot{\theta} = J^\top q(r)/n$ yields $\dot{r} = JJ^\top q(r)/n$ and $\dot{M}_a = c_a^\top q(r)/n$, proving Eq. (10). Under exact response-coordinate realization, $\nabla r_i = \sum_b z_{ib} \nabla M_b$, and substitution yields Eq. (11). If $r_i = z_i \cdot M + r_i^\perp$, the exact residual is

$$R_a = \frac{1}{n} \sum_i \sigma(-r_i) \langle \nabla M_a, \nabla r_i^\perp \rangle, \quad \dot{M}_a = (Gg)_a + R_a. \quad (22)$$

A.2 PROOF OF DENSE-LINEAR GEOMETRY

For $M_a = c^\top W^{\ell_a} b_a$, differentiation with respect to c gives $p_a = W^{\ell_a} b_a$, differentiation with respect to the channel- a input column gives $q_a = (W^{\ell_a})^\top c$, and differentiation with respect to W gives

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (23)$$

The parameter blocks are orthogonal in the implemented product Euclidean/Frobenius metric. Their gradient inner products add, and input-column terms contribute only for a shared channel, giving Eq. (12).

A.3 PROOF OF PROPOSITION 2 AND COROLLARY 3

For every real x , $x = [x]_+ - [-x]_+$. Absolute continuity therefore gives

$$\Delta(t) - \Delta(0) = \int_0^t d(u) du = \int_0^t [d(u)]_+ du - \int_0^t [-d(u)]_+ du = P(t) - N(t). \quad (24)$$

Since $\Delta(0) = 0$, Eq. (15) follows. Strict outcome suppression occurs at time t exactly when $N(t) > P(t)$; taking existence over $t \leq H$ proves the stated criterion for arbitrary sign changes.

Under the one-crossing hypothesis, $P(t) = A_+$ for $t \geq \tau_d$ and $N(t) = N_{\text{tail}}(t)$, hence $\Delta(t) = A_+ - N_{\text{tail}}(t)$. This gives Corollary 3 immediately. If $d(t) \leq -\kappa < 0$ after $t_0 > \tau_d$, then $N_{\text{tail}}(t) - N_{\text{tail}}(t_0) \geq \kappa(t - t_0)$, so

$$\tau_{\text{response}} \leq t_0 + \frac{[A_+ - N_{\text{tail}}(t_0)]_+}{\kappa} \quad (25)$$

whenever the right side lies within the horizon.

A.4 INITIALIZATION-LOCAL SUFFICIENT CERTIFICATE

The current implementation computes the exact paired initialization jet under the full-batch CE vector field: $\Delta(0)$, $d_0 = \Delta'(0)$, and $j_0 = \Delta''(0)$, without taking an optimization step. The following result is a rigorous local sufficient certificate; it is not used as an empirically calibrated predictor in the main paper.

Proposition 5 (Initial-jet local onset certificate). *Assume $\Delta \in C^2([0, H])$, $\Delta(0) = 0$, and write $d = \Delta'$, $j = d'$. Suppose $d_0 = d(0) > 0$, $j_0 = j(0) < 0$, and j is L -Lipschitz on $[0, H]$. Let*

$$h_* = \min \left\{ H, \frac{-j_0}{2L} \right\}, \quad (26)$$

with $-j_0/(2L) = +\infty$ when $L = 0$. Then on $[0, h_*]$,

$$d(t) \leq d_0 - \frac{-j_0}{2} t, \quad \Delta(t) \leq d_0 t - \frac{-j_0}{4} t^2. \quad (27)$$

Consequently: (i) if $h_* > 2d_0/(-j_0)$, a transfer-to-suppression rate crossing occurs by $2d_0/(-j_0)$; (ii) if $h_* > 4d_0/(-j_0)$, then $\Delta(t) < 0$ for some $t \leq h_*$, so outcome suppression occurs. If the weak-only learnability gate also holds by that time, causal starvation follows.

Proof. Lipschitzness implies $j(t) \leq j_0 + Lt \leq j_0/2 < 0$ for $t \leq h_*$. Integrating once gives the bound on d , and integrating again with $\Delta(0) = 0$ gives Eq. (27). The first upper bound is nonpositive by $2d_0/(-j_0)$; continuity gives a rate crossing, and strict negativity of j makes it unique on this interval. The second upper bound is strictly negative for $t > 4d_0/(-j_0)$, proving outcome suppression. $\square$

A convenient sufficient form is $H > 4d_0/(-j_0)$ together with $j_0^2 > 8Ld_0$. The proposition shows what an initialization-only certificate would require; obtaining a useful, non-vacuous L bound for broad nonlinear RNNs remains separate work.

A.5 TRANSVERSE FIRST-HIT STABILITY

Let continuous $x_N, x : [0, H] \rightarrow \mathbb{R}$ satisfy $\|x_N - x\|_\infty \leq \epsilon$, with target β and reference first hit $\tau \in (0, H)$. Assume $x'(t) \geq \kappa > 0$ almost everywhere on $[\tau - r, \tau + r]$ and a positive prehistory separation from β . If ϵ is smaller than that separation and $\epsilon/\kappa \leq r$, then $|\tau_N - \tau| \leq \epsilon/\kappa$. On a grid of maximum spacing h , the sampled/interpolated error is at most $\epsilon/\kappa + h$. Applying the result to both conditions and using the triangle inequality gives the paired hitting-time delay bound.

B PROOF OF CDC GUARANTEES

If $a = 0$, strong drift vanishes. Otherwise Eq. (18) gives $a^\top q = 0$, hence $a^\top v' = a^\top v$. For minimum norm, let $\delta > 0$. Every feasible d satisfies $a^\top d = 0$ and $p^\top d \geq \delta$. Because $p = q + (p^\top a / \|a\|^2)a$, feasibility implies $q^\top d \geq \delta$. Cauchy-Schwarz gives $\|d\| \geq \delta/\|q\|$, with equality only for the nonnegative multiple $d^* = (\delta/\|q\|^2)q$. Thus d^* is the unique norm-minimal feasible correction.

If ∇M_s is L -Lipschitz on both straight coordinate segments, smoothness gives

$$|M_s(\theta + h) - M_s(\theta) - \nabla M_s(\theta)^\top h| \leq \frac{L}{2} \|h\|^2. \quad (28)$$

Apply this at $h = \eta v$ and $h = \eta v'$. The linear terms cancel by instantaneous preservation; the triangle inequality yields Eq. (19).

C ADDITIONAL EXPERIMENTAL DETAILS

C.1 DENSE-LINEAR PROTOCOL

The dense-linear causal grid uses width 96, $\rho \in \{1, 2, 4, 8\}$, lag $\in \{0, 2, 4, 8\}$, 256 samples, 500 full-batch steps, learning rate 0.01, recurrent gain 0.5, target $\beta = 0.25$, and eight paired seeds. The reversed-cue negative control is evaluated at $(\rho, \text{lag}) = (4, 4)$. Hitting times beyond the training horizon are right-censored rather than converted to finite delays.

C.2 NONLINEAR PROTOCOL AND THRESHOLD SCOPE

The nonlinear audit uses tanh RNNs and GRUs of width 32, $\rho = 4$, lag 2, 256 samples, 1000 full-batch steps, learning rate 0.01, cue noise 0.1, zero background noise, $\beta = 0.5$, and eight paired seeds. Both and weak-only models are advanced in lockstep so the exact direct-autograd drift difference is available at each logged point. The β -dependent counts in Table 1 are therefore operating-point statements. The threshold-free response-crossing counts are reported separately and should not be interpreted as causal-starvation counts.

C.3 DISCRETE ONE-CROSSING CERTIFICATE

For logged gaps Δ_k , increments telescope exactly. The preregistered executable certificate used for the nonlinear table requires one strict block of positive increments followed by one strict block of negative increments. Peak-minus-initial is the discrete positive area and peak-minus-final is the negative tail area. This certificate is intentionally stricter than Proposition 2; trajectories with multiple sign changes or immediate suppression are not labeled by the one-crossing routine.

D FULL CDC ABLATION

Table 3: Eight-method dense-linear mitigation ablation, eight paired seeds. “Strong change” is the maximum absolute instantaneous strong-drift perturbation.

Method	weak AUC gap	strong change	final M_s
ERM	10.5155	—	1.364
Spectral Decoupling	9.0229	—	0.829
Interaction penalty	10.5145	—	1.365
CDC	−0.0005	4.77×10^{-7}	1.042
Loss-gradient projection	−0.0002	3.75×10^{-1}	0.371
Unconstrained rescue	−0.0007	1.48×10^{-1}	1.295
Bloop-style	0.0002	3.75×10^{-1}	0.370
PCGrad-style	−0.0007	1.48×10^{-1}	1.295

All five shadow-based methods attain the same weak-drift target at every logged step and produce weak outcomes that are numerically close relative to the roughly 10.5-point change from ERM. CDC’s established distinction is its exact local strong-drift constraint, not superior final strong retention or uniquely better weak rescue.

E NEGATIVE AND SCOPE AUDITS

An initialization-frozen tangent surrogate was evaluated in a separate sealed study. A broad preregistered rule that included phase, learnability, and causal labels failed; a later fresh restricted study classified drift/response crossing events well at one controlled task cell but did not calibrate crossing times or trajectory magnitudes. We therefore exclude this surrogate from the main claims. This negative result is retained to prevent post-hoc promotion of a partially successful proxy into a general nonlinear theory.

The legacy five-index GSI pipeline is likewise not used as a theorem. GSI-5 is a concentration statistic of the CE weights $q(r)$ and remains a useful mechanism diagnostic, but causal weak drift is $c_w^\top q/n$ and additionally depends on the matched counterfactual and pathwise response budget.

F REPRODUCIBILITY AND SELECTION AUDIT

The final reported dense-linear experiments use the corrected width-scaled input parameterization and final response definitions. Earlier exploratory runs are excluded wholesale because they used either duplicated strong-feature scaling or a superseded nonlinear response convention. Those runs therefore measured a different mathematical object from the final protocol; the paper does not combine or

648 selectively reuse their outcomes across definitions. The anonymous supplementary package should
649 contain the theorem-aligned source, all resolved configurations, tests, citable result tables, and a
650 machine-readable provenance record. No public repository URL, commit hash, or tag should appear
651 in the double-blind submission.
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